5 (a) Complete the table of values for $y = x^2 - x - 6$

x	–3	–2	–1	0	1	2	3
y	6	0	-4	– 6	-6	-4	0

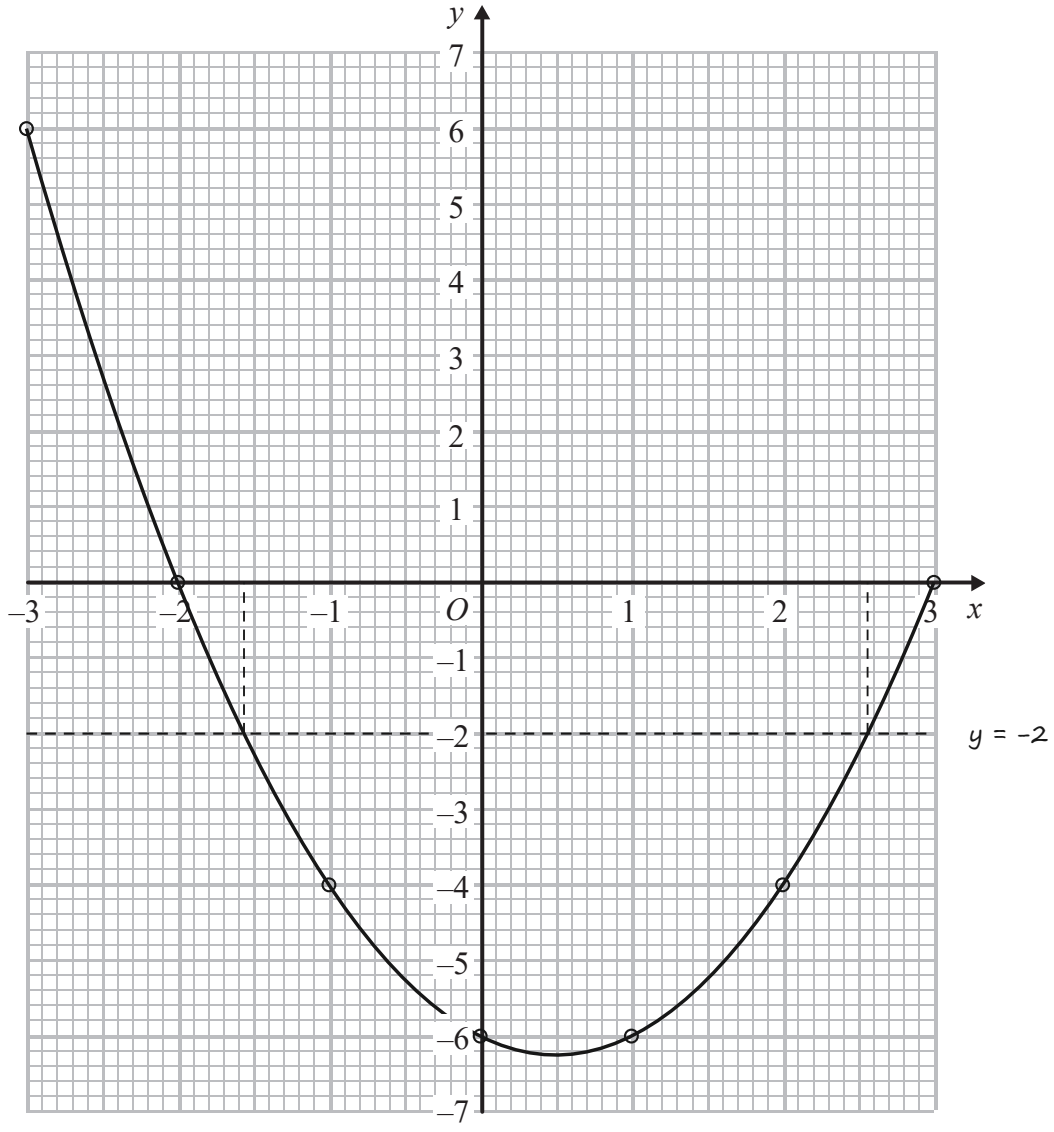
For example, when x = -2: $y = (-2)^2 - (-2) - 6 = 0$.

2 marks total
All five missing values correct

(2)

(b) On the grid, draw the graph of $y = x^2 - x - 6$ for values of x from –3 to 3

(2)



1 mark
Plot at least five correct points

1 mark
Join with a smooth curve

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(c) Use your graph to find estimates of the solutions to the equation $x^2 - x - 6 = -2$

Draw y = -2 and read the x-coordinates where it meets the curve.

1 mark
Use the horizontal comparison line

1 mark
Both estimates correct (within reading tolerance)

-1.6 and 2.6

(2)

(Total for Question 5 is 6 marks)

6 A force of 70 newtons acts on an area of 20 cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The force is increased by 10 newtons.
The area is increased by 10 cm^2

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?
You must show how you get your answer.

(Total for Question 6 is 3 marks)

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